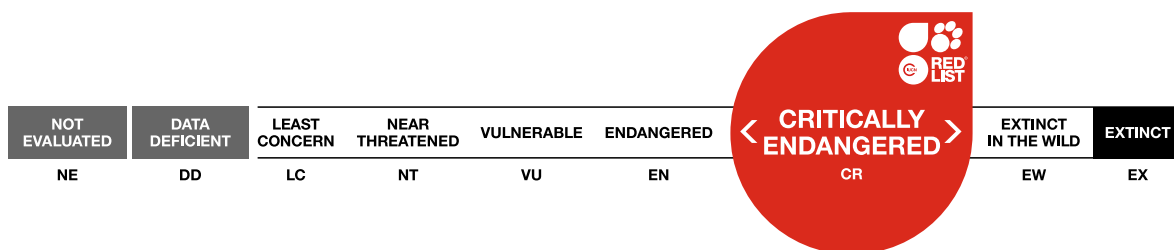


Xenopus longipes, Lake Oku Clawed Frog

Amendment version

Assessment by: IUCN SSC Amphibian Specialist Group



View on www.iucnredlist.org

Citation: IUCN SSC Amphibian Specialist Group. 2020. *Xenopus longipes* (amended version of 2017 assessment). *The IUCN Red List of Threatened Species* 2020: e.T58176A177346697.

<https://dx.doi.org/10.2305/IUCN.UK.2020-3.RLTS.T58176A177346697.en>

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Taxonomy

Kingdom	Phylum	Class	Order	Family
Animalia	Chordata	Amphibia	Anura	Pipidae

Scientific Name: *Xenopus longipes* Loumont & Kobel, 1991

Common Name(s):

- English: Lake Oku Clawed Frog, Savanna Clawed Frog

Taxonomic Source(s):

Frost, D.R. 2016. Amphibian Species of the World: an Online Reference. Version 6.0 (31 March 2016). New York, USA. Available at: <http://research.amnh.org/herpetology/amphibia/index.html>.

Taxonomic Notes:

Uniquely among vertebrates (except *Xenopus eysoole*, *X. kobeli* and *X. ruwenzoriensis*), this is a dodecaploid species, and it is therefore of considerable scientific interest. It was probably formed by both hybridization and polyploidization (Loumont and Kobel 1991).

Assessment Information

Red List Category & Criteria: Critically Endangered B1ab(iii,v)+2ab(iii,v) [ver 3.1](#)

Year Published: 2020

Date Assessed: February 6, 2017

Justification:

Listed as Critically Endangered because its extent of occurrence (EOO) and area of occupancy (AOO) is 2 km², it occurs in one threat-defined location, and there is a projected decline in the number of mature individuals, due to the high likelihood of a fish introduction into Lake Oku, habitat degradation which is causing subsequent decline in the quality of the water in the lake, and perhaps the compounded effects of disease.

Previously Published Red List Assessments

2017 – Critically Endangered (CR)
<https://dx.doi.org/10.2305/IUCN.UK.2017-2.RLTS.T58176A3067255.en>

2004 – Critically Endangered (CR)
<https://dx.doi.org/10.2305/IUCN.UK.2004.RLTS.T58176A11731190.en>

Geographic Range

Range Description:

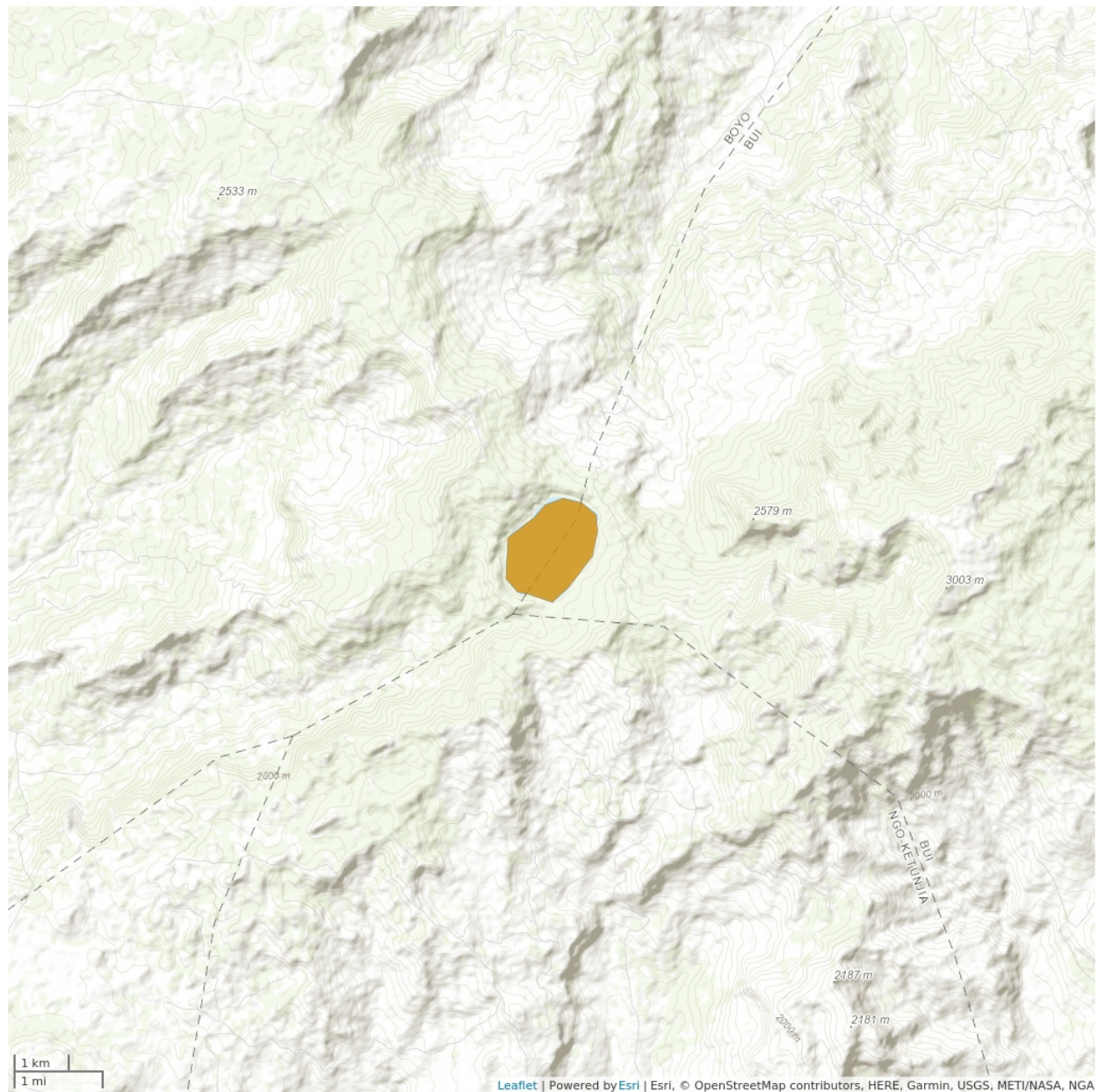
This species is endemic to Lake Oku situated at 2,200 m Asl on Mount Oku in western Cameroon. While it might occur in other crater lakes in the Cameroon highlands, it was not recorded in these lakes during surveys in 2012 (T. Doherty-Bone pers. comm. July 2012). The area of the lake is only 243 hectares (Kling

1988). Its EOO and AOO is 2 km².

Country Occurrence:

Native, Extant (resident): Cameroon

Distribution Map

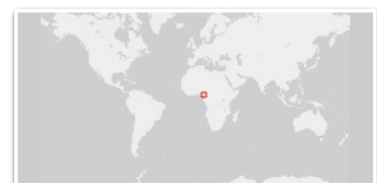


Legend

EXTANT (RESIDENT)

Compiled by:

IUCN (International Union for Conservation of Nature) & Conservation International. 2017



The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.

Population

This species is abundant in Lake Oku. However, since 2006, recurring mass mortality events have been recorded. These remain enigmatic and might be due to disease, although testing for *Bd* and *Ranavirus* has been negative on multiple occasions or of low prevalence (Blackburn *et al.* 2010, Doherty-Bone *et al.* 2013, Quock *et al.* 2014, Hirschfeld *et al.* 2016). Recent surveys still suggest that the species occurs in large numbers (T. Doherty-Bone unpubl. data).

Current Population Trend: Decreasing

Habitat and Ecology (see Appendix for additional information)

Lake Oku is a shallow, oligotrophic lake completely surrounded by montane rainforest. This species is entirely water-dependent, quite inept on land, and is the main aquatic vertebrate in the lake, likely filling an ecological niche similar to an omnivorous or insectivorous fish.

Breeding ecology is not clear as breeding behaviour has not been consistently observed, tadpoles are rarely observed alive in the lake and adults are the same size as metamorphs (T. Doherty-Bone pers. comm. February 2017). When tadpoles have been found washed up at lake shore, this has been between July and October. In captivity, the larvae took six months and longer to metamorphose (Michaels *et al.* 2015).

Systems: Freshwater (=Inland waters)

Use and Trade

There are no records of this species being utilized. Experts were concerned about the increasing international interest in this species and worried the species would be harvested for international trade in the near future, if it is not already occurring. However, this is based on one incident of an animal dealer contacting a Cameroonian field assistant for collection, which turned out to be merely prospective (T. Doherty-Bone pers. comm. February 2017).

Threats (see Appendix for additional information)

One potential threat to this species would appear to be the risk of introduction of fish into Lake Oku, which could wipe out this restricted-range species. Such a scenario is highly plausible given the high protein demands of surrounding communities. However there have been no official fish introductions into the lake as of yet, and fish introduction continues to be blocked by the local forestry and wildlife office (which manages the protected area encompassing the lake) and opposed by traditional authorities (Fon and elders) and local environmentalists who also encourage the abstinence of fish introduction (T. Doherty-Bone pers. comm. November 2016). Surveys between 2006–2012 documented an unknown disease associated with mass mortality events, possibly involving a pathogen (Doherty-Bone *et al.* 2013).

Another threat includes the degradation of the forest around Lake Oku, which could alter the water quality through siltation and nutrient pollution. This occurs through livestock grazing and forest loss due to wood cutting and infrastructure developments. These are likely to be exacerbated by increases in visitors to the lake as a result of these developments, and if unregulated could induce increased

demands for fuel wood and garbage accumulation (T. Doherty-Bone pers. comm. February 2017). There have also been cases when visitors to the lake have washed clothes using chemical detergents. Loss of forest around the lake could also increase run-off of chemicals from vehicles on the road above the lake (T. Doherty-Bone pers. comm. February 2017).

A retrospective study analysing amphibian population declines (between 2004–2012) and *Bd* emergence (in 2008) on Mount Manengouba and Mount Oku in Cameroon suggest that chytridiomycosis has driven community level declines of anuran biodiversity in these hotspot areas (Hirschfeld *et al.* 2016). This species was not tested for *Bd* during Hirschfeld *et al.* (2016) study so it is not known whether chytridiomycosis is a threat, but other species in the genus had mixed results.

Conservation Actions (see Appendix for additional information)

Conservation Actions

Lake Oku and its surrounding forest are protected by the national government-sanctioned Kilum-Ijim Plantlife Sanctuary, restricting access to the lake; though enforcement is difficult and even controversial. In response to the 2004 assessment of *Xenopus longipes*, a captive-breeding programme is being attempted by European zoos (London, Antwerp and Cologne) in view of the risk of a catastrophic collapse of the population (Browne *et al.* 2009). The first successful captive breeding of this species was achieved by ZSL London Zoo in 2014 (Michaels *et al.* 2015). A conservation project was also conducted on Mount Oku for several years by BirdLife International, which involved community management of mountain forest involving the local villages. However, the project ended in the mid-2000s (N. Gonwouo pers. comm. June 2012).

The mysterious disease that has affected this species has been monitored since its discovery in 2006. However, its cause and long-term consequences are still not understood other than its apparent stability amid a numerically abundant frog population (Doherty-Bone *et al.* 2013). While chytrid is present in the region, it is apparently not affecting this species and is not the culprit of the observed die-offs. Since 2008, a project has monitored the abundance of frogs, occurrence of disease, fluctuations of environmental parameters in the lake, and raised awareness and engagement with local communities (Doherty-Bone 2015). The lake is situated just outside the boundary of a proposed Faunal Reserve on Mount Oku.

Conservation Needed

A review and update of the 2013 conservation action plan is due (T. Doherty-Bone pers. comm. February 2017). Future actions will likely include the continuation of disease monitoring, increasing community involvement and awareness, the continuation of ex-situ breeding programmes, and improving the protection of Lake Oku (prevention of fish stocking) and its surrounding forest. Recent development above the lake also indicate engagement with those stakeholders (from outside the region) is necessary to minimise current and future developments that could over-rule local decisions for protection (T. Doherty-Bone pers. comm. February 2017).

Research Needed

Research is required to establish the cause of the observed die-offs, to determine the spatial distribution of the frog across the lake and estimate of minimum viable populations, to understand the broader ecology of Lake Oku (such as food webs and submerged vegetation), to understand the reproductive ecology of this species, to see whether there is the potential for other *Xenopus* species to colonise Lake Oku and out-compete this species, for socio-economic appraisals of community and opportunities for sustainable protection of the lake, to establish the effect by *Bd* infection on this species, and to determine the effectiveness of conservation interventions (T. Doherty-Bone pers. comm. February 2017). There is also the need to remain vigilant to harvest for international trade (T. Doherty-Bone pers. comm. February 2017).

Credits

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Authority/Authorities: IUCN SSC Amphibian Specialist Group

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Citation

IUCN SSC Amphibian Specialist Group. 2020. *Xenopus longipes* (amended version of 2017 assessment). *The IUCN Red List of Threatened Species* 2020: e.T58176A177346697.

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External Resources

For [Supplementary Material](#), and for [Images and External Links to Additional Information](#), please see the Red List website.

Appendix

Habitats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Habitat	Season	Suitability	Major Importance?
5. Wetlands (inland) -> 5.5. Wetlands (inland) - Permanent Freshwater Lakes (over 8ha)	Resident	Suitable	Yes

Use and Trade

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

End Use	Local	National	International
Pets/display animals, horticulture	No	No	No

Threats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Threat	Timing	Scope	Severity	Impact Score
1. Residential & commercial development -> 1.1. Housing & urban areas	Ongoing	-	-	Low impact: 3
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
1. Residential & commercial development -> 1.3. Tourism & recreation areas	Ongoing	-	-	Low impact: 3
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
2. Agriculture & aquaculture -> 2.3. Livestock farming & ranching -> 2.3.2. Small-holder grazing, ranching or farming	Ongoing	-	-	Low impact: 3
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation		
5. Biological resource use -> 5.3. Logging & wood harvesting -> 5.3.3. Unintentional effects: (subsistence/small scale) [harvest]	Ongoing	-	-	Low impact: 3
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation		
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.1. Unspecified species	Future	Whole (>90%)	Unknown	Unknown
	Stresses:	2. Species Stresses -> 2.1. Species mortality 2. Species Stresses -> 2.2. Species disturbance		

8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (Batrachochytrium dendrobatidis)	Ongoing	Unknown	Unknown	Unknown
	Stresses:	2. Species Stresses -> 2.1. Species mortality		
8. Invasive and other problematic species, genes & diseases -> 8.6. Diseases of unknown cause	Past, likely to return	Majority (50-90%)	-	Past impact
	Stresses:	2. Species Stresses -> 2.1. Species mortality		

Conservation Actions in Place

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Action in Place
In-place land/water protection
Area based regional management plan: Yes
Occurs in at least one protected area: Yes

Conservation Actions Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Action Needed
1. Land/water protection -> 1.1. Site/area protection
1. Land/water protection -> 1.2. Resource & habitat protection
2. Land/water management -> 2.2. Invasive/problematic species control
3. Species management -> 3.4. Ex-situ conservation -> 3.4.1. Captive breeding/artificial propagation
4. Education & awareness -> 4.3. Awareness & communications

Research Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Research Needed
1. Research -> 1.2. Population size, distribution & trends
1. Research -> 1.3. Life history & ecology
1. Research -> 1.4. Harvest, use & livelihoods
1. Research -> 1.5. Threats
3. Monitoring -> 3.2. Harvest level trends

Additional Data Fields

Distribution
Estimated area of occupancy (AOO) (km ²): 2.43
Estimated extent of occurrence (EOO) (km ²): 2.15
Number of Locations: 1
Lower elevation limit (m): 2,200
Upper elevation limit (m): 2,200
Population
Continuing decline of mature individuals: Yes
Population severely fragmented: No
Habitats and Ecology
Continuing decline in area, extent and/or quality of habitat: Yes
Movement patterns: Not a Migrant

Amendment

Amendment reason: Correction of coded threats: logging and wood harvesting has unintentional effects on the species, rather than the species being intentionally used for this purpose.

The IUCN Red List Partnership



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